



Brain computer interface (photo courtesy: ARS Electronica)

(EEG), magnetoencephalography (MEG) and functional near-infrared spectroscopy (fNIRS). These techniques are non-invasive ways of studying brain waves because they do not involve surgery of the patient. Electroencephalography (EEG) is another technique to read neural activities which is invasive.

These techniques generate huge data sets that are analysed for different features of neural signaling. Neural signals are essentially pulse streams. Research is underway on the ways to comprehend such pulse-train patterns. Various approaches like pattern recognition and adaptive signal processing algorithms are being deployed to classify and characterise the signals. Artificial neural network programming using Matlab is very popular in extracting the features of neural signals. Various statistical methods are employed in devising the algorithms like principle component analysis and independent component analysis that are used to classify neural signals and use them to control the device operations.

2. Neuron modeling. Development of behavioural models of neurons is necessary for experimental and research work. Mathematical models of neurons are developed keeping the utility of neurons in mind. (The need is similar to development of CMOS transistor models in the field of semiconductor.) Because of our exemplary understanding of electrical models, electrical

probes have to have suitable properties depending upon whether invasive or non-invasive technique is employed. In the invasive technique, probes are directly implanted into the brain to connect to the neurons. The non-invasive technique lets the probes to be fixed on the outer surface of the skull. Development of very sensitive probes is essential for non-invasive technique, while tissue-friendly probes are more important for invasive techniques.

Major areas of research for developing the right probes are biocompatibility, high-sensitivity and high-selectivity material development for probes, and reliability:

(i) **Biocompatibility.** Neuron-material interfacing is the foremost hurdle in the development of the invasive BCI. The search for the ideal probe material that can be directly linked to the neurons in the brain to be tightly coupled with the neurons is essential to collect the baseband signal. Various techniques to convert bio-inert material into bio-mimetic material are under research. Surface functionalisation and plasma processing are some of the techniques employed to convert substrate materials like silicon into bio-mimetic material by introducing a functional group chemically into the structure.

Reliability of the implantable materials is a major concern. Chemical reactions due to various fluids and environmental factors of the brain can corrode the material. Therefore a

careful search is essential. Besides, a neuron is several micrometres in diameter, while the size of the conventional probes is tens of micrometres. This mismatch in size can damage a neuron during penetration of the probes into the brain. Therefore smaller size of probe is pertinent.

(ii) **High-sensitivity and high-selectivity material development for probes.** Usually, neural signals are captured through biochemical, optical or electromagnetic means. The high density of neurons makes placing the probe at the right location very difficult. Closely packed neurons also induce noise at the inputs of the probes and thus increase chances of error in deciphering the correct signal. Therefore the probes need to have high noise immunity and selectivity so that only the neural signal of interest is selected.

(iii) **Reliability.** The reliability of probes over time is a very pressing issue. Following fundamental problems of implantable microelectrode arrays are to be fixed so that probes can capture signals reliably over time:

(a) The brain often encapsulates the device with a scar tissue
(b) Normal brain movement may cause micro-motion at the tissue-electrode interface
(c) Proteins adsorb onto the device surface
(d) Useful neural recordings are therefore lost over time

Neural processor. The signal extractor collects electrical signals from neural probes and processes them to extract the signal information to be used to control the various devices interfaced to the BCI system. A low-noise preamplifier refines the neural signals detected by probes. Its output has a high signal-to-noise ratio and can be fed to a high-speed analogue-to-digital converter (ADC) so that the signals can be processed using a neural processor.

The neural processor is an ASIC or a DSP tuned for neuro-computing.

Advancement in the VLSI technology is a major factor in realisation of BCI systems. The processor has